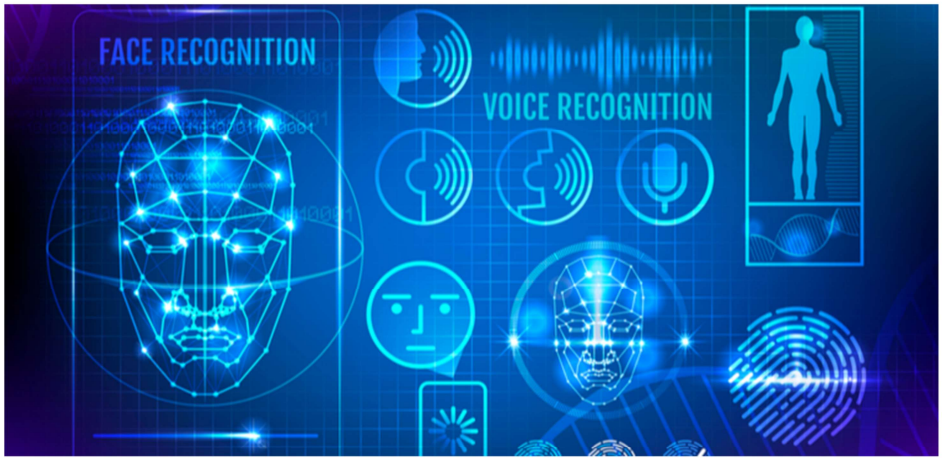
Biometrics



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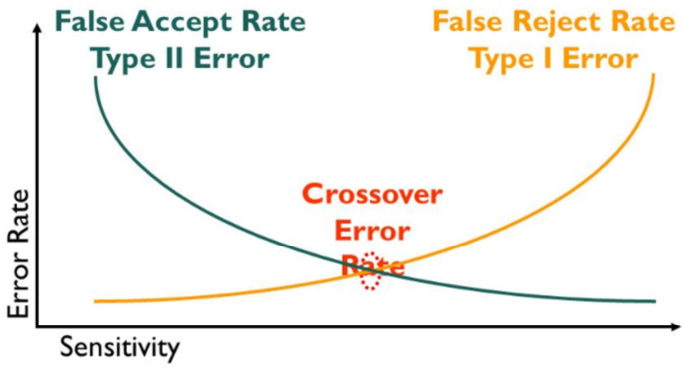
Biometric devices rely on measurements of biological characteristics of an individual, such as a fingerprint, hand geometry, voice, or iris patterns. Biometric technology involves data that is unique to the individual and is difficult to counterfeit. Selected individual characteristics are stored in a device's memory, or on a card, which stores reference data that can be analyzed and compared with the presented template. A one-to-many or a one-to-one comparison of the presented template with the stored template can be made and access granted if a match is found. However, on the negative side, some biometric systems may periodically fail to perform, or have a high rejection rate. The sensitivity of readers makes system readers susceptible to inadvertent reader damage or intentional sabotage. Some systems may be perceived by the user as a safety or health risk. Also, some of the systems may require a degree of skill on the part of the user for proper operation. Other systems may be perceived as unacceptable by management for a combination of reasons.

Types of Failure in Biometric Identification

- False Rejection Rate (Type I)
- False Acceptance Rate (Type II)

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False Rejection Rate (Type I) vs False Acceptance Rate (Type II)



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Types of Failure in Biometric Identification
There are two types of failures in biometric identification:

False Rejection Rate (Type I): This is a failure to recognize a legitimate user. While it could be argued that this effectively keeps the protected area extra secure, it is an intolerable frustration to legitimate users who are refused access because the scanner does not recognize them.

False Acceptance Rate (Type II): This is erroneous recognition, either by confusing one user with another, or by accepting an imposter as a legitimate user. Failure rates can be adjusted by changing the criteria for declaring an acceptance or rejection; but decreasing one failure rate increases the other. Crossover Error Rate (CER) is achieved when the type I and type II are equal.

Biometric Readers

- Fingerprint
- Facial image
- Hand geometry
- Voice recognition
- Iris patterns
- Retinal scanning
- Signature dynamics
- Vascular patterns
- Keystroke dymanics

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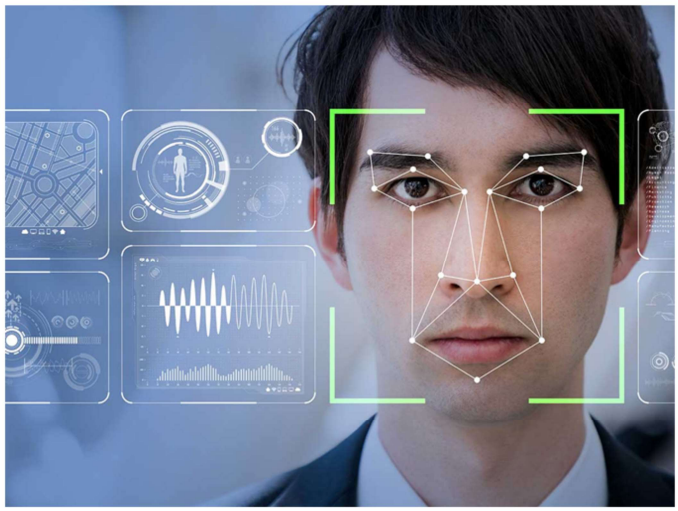
Fingerprint



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Fingerprint: Fingerprint reader technology scans the loops, whorls, and other characteristics of a fingerprint and compares it with stored templates. When a match is found, access is granted. The advantage of fingerprint technology is that it is easily understood. The disadvantages are that the system can be disrupted if cuts or sores appear on fingers, or if grease or other medium contaminates the fingers and the scanning plates.

Facial image

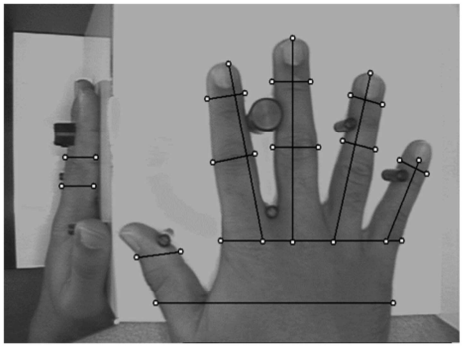


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Facial image: This technology measures the geometric properties of the subject's face relative to an archived image. Specifically, the center of the subject's eyes must be located and placed at precise locations.

Hand geometry

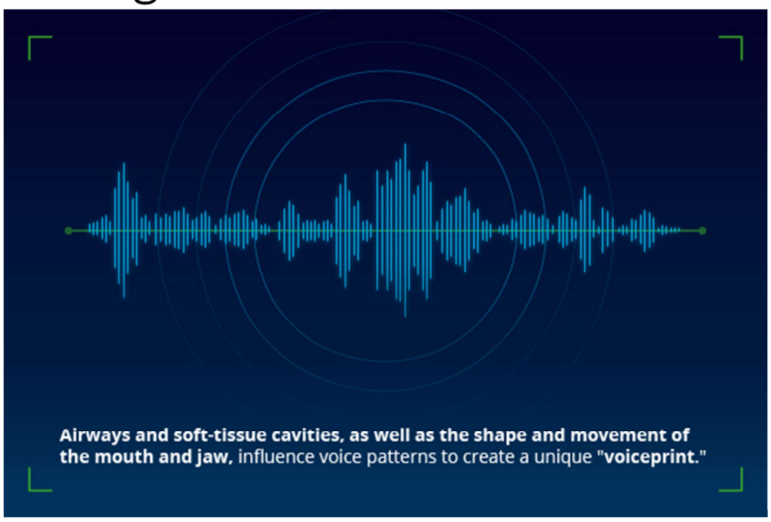




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Hand geometry: This technology assesses the hand's geometry: height, width, and distance between knuckle joints and finger length. Advantages of hand geometry are that the systems are durable and easily understood. The speed of hand recognition tends to be more rapid than fingerprint recognition. Hand recognition is reasonably accurate because the shape of a hand is unique. A disadvantage is that hand recognition tends to give higher false acceptance rates than fingerprint recognition.

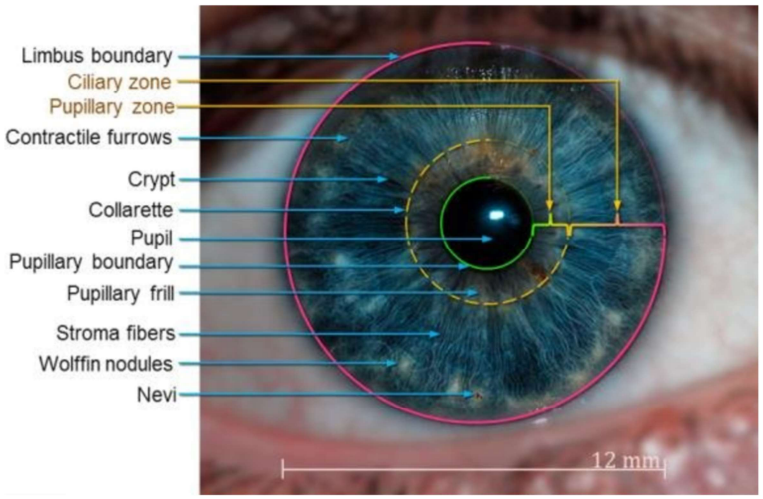
Voice recognition



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Voice recognition: Voice recognition compares the voice characteristics of a given phrase to one held in a template. Voice recognition is generally not performed as one function and is typically part of a system where a valid PIN must be entered before the voice analyzer is activated. Advantages of voice recognition are that the technology is less expensive than other biometric technologies, and it has hands-free operation. A disadvantage is that the voice synthesizer must be placed in an area where the voice is not disturbed by background sounds; often a booth or security portal must be installed to house the sensor to provide the system with a quiet background.

Iris patterns

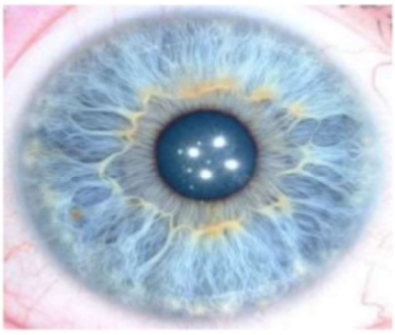


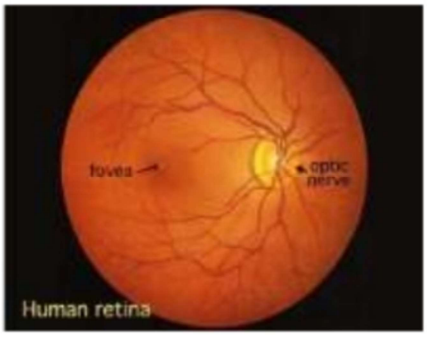
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Iris patterns: Iris recognition technology scans the surface of the eye and compares the iris pattern with stored iris templates. An advantage of iris recognition is that it is not susceptible to theft, loss, or compromise, and irises are less susceptible to wear and injury than many other parts of the body. Newer iris scanners allow scanning to occur from up to ten inches away. A disadvantage of iris scanning is that some people are timid about having their eye scanned. Throughput time for this technology also should be considered; typical throughput time is two seconds. If a number of people need to be processed through an entrance in a short period of time, this can be problematic.

Retinal scanning

Iris scanning	Retina scanning

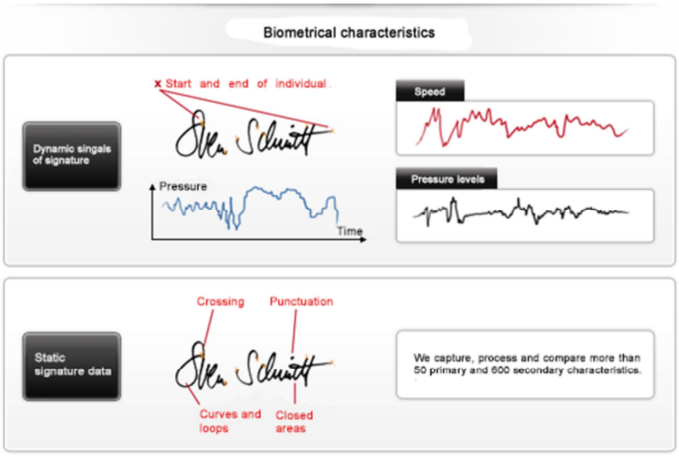




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Retinal scanning: Retinal scanning analyzes the layer of blood vessels at the back of the eye, which are unique to each person. Scanning involves using a low-intensity LED light source and an optical coupler that can read the patterns with great accuracy. It does require the user to remove glasses, place the eye close to the device, and focus on a certain point. The user looks through a small opening in the device, and the head needs to be still and the eye focused for several seconds, during which time the device verifies identity. This process takes about ten seconds. The continuity of the retinal pattern throughout life and the difficulty in fooling such a device also makes it a great long-term, high-security option.

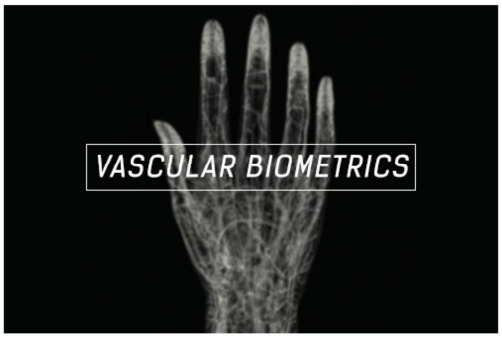
Signature dynamics biometric

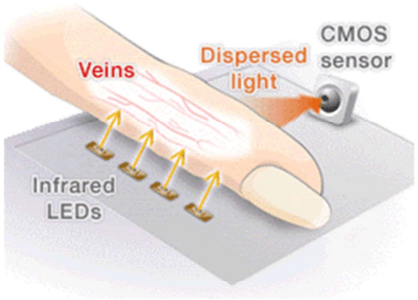


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Signature dynamics: First, the signer writes out a handwritten signature on a special electronic pad, such as the ePad by Interlink or a Palm Pilot. The shape of the signature is then electronically read and recorded, along with unique features, such as the pressure on the pen and the speed at which the signature was written, to identify the signer's unique writing; for example, did the "t" get crossed from right to left and did the "i" get dotted at the very end. The advantage of signature dynamics is that it works like a traditional signature. Signers do not need special knowledge of computers nor any unusual tools to provide a signature. At the same time, the system allows the notary to record unique identifying features to help prevent and detect forged signatures.

Vascular patterns biometric

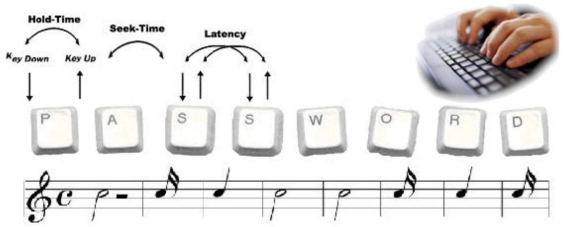


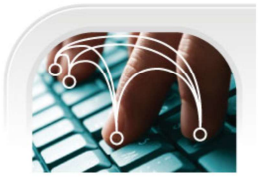


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Vascular patterns: This is the ultimate palm reader; vascular patterns are best described as a picture of the veins in a person's hand or finger. The thickness and location of these veins are believed to be unique enough to an individual to verify a person's identity. The National Television Standards Committee (NTSC) Subcommittee on Biometrics reports that researchers determined that the vascular pattern of the human body is unique to each individual and does not change with age.

Keystroke dynamics





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Keystroke dynamics: Keystroke dynamics are also known as keyboard dynamics, which identify the way a person types at a keyboard; specifically, the keystroke rhythms of a user are measured to develop a unique template of the user's typing pattern for future authentication. Raw measurements available from most keyboards can be recorded to determine dwell time, or the amount of time a particular key is held, and flight time, or the amount of time between the next key down and the next key up.

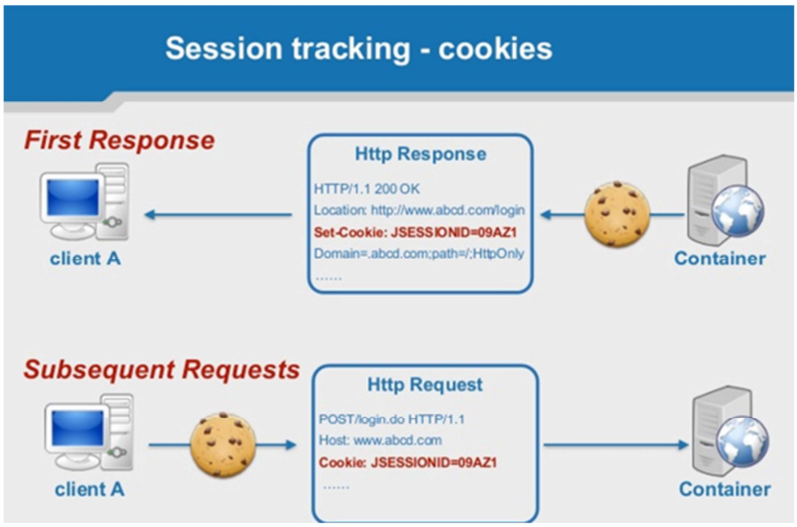
Authorization



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Authorization defines what resources users may have access to.

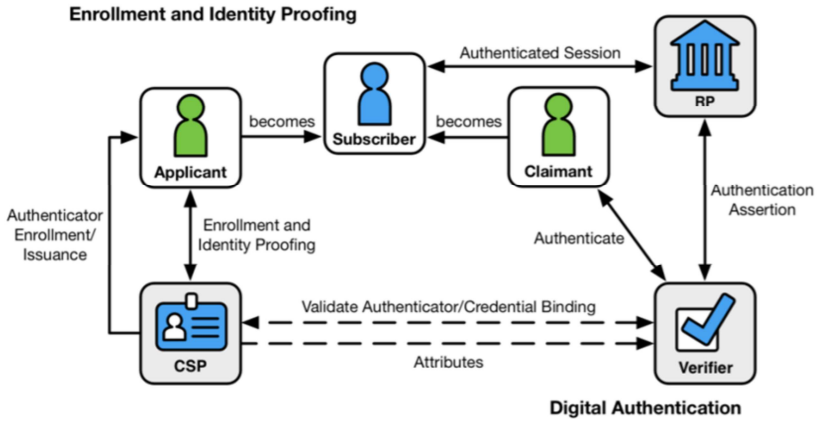
Session Management



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Session management is related to when a user is authenticated, authorized, and held accountable for using system resources. The system must maintain an uninterrupted path of protection of resources by means of system management. Open Web Application Security Project (OWASP) Top 10 number 2 threat is broken authentication and session management. RFC 2965 provides an example of how to maintain session managements with cookies. When a user accesses a website, the user's actions and identity are tracked across various requests from that website. A state of these interactions is maintained in a session cookie. Evidence of this state is maintained by linking all new connections across the entirety of a session to the cookie. Cookie handling achieves non-repudiation; effectively leveraging an audit trail of session activity.

Registration and Proofing of Identity



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Registration and proofing of an identity are processes that connect an entity or user identity to an access control system that creates a confirmed relationship of trust that an entity is who he or she claims to be. The process of proving that a person is authentically the person that is being claimed can be challenging and even serve as an opening for impersonation. If a user is valid, there is also the threat that the user can be a malicious or bad actor. Writing for the *New Yorker*, Peter Steiner stated succinctly, "On the Internet no one knows that you are a dog."

Herein lies the crux of the concern; balancing the needs of controlling access to valued assets and the simplicity of registering and proofing the credentials of the potential user of a system.

The Digital Identity Guidelines of NIST SP 800-63-3 contains recommendations to support, among other items, requirements for identity proofing and registration. These requirements are the following:

Identity Assurance Level (IAL) refers to the identity proofing process. A category that conveys the degree of confidence that the applicant's claimed identity is their real identity

Identity Assurance Levels

IAL1: At IAL1, attributes, if any, are self-asserted or should be treated as self-asserted.

IAL2: At IAL2, either remote or in-person identity proofing is required.

IAL2 requires identifying attributes to have been verified in person or remotely, using, at a minimum, the procedures given in SP 800-63A.

IAL3: At IAL3, in-person identity proofing is required. Identifying attributes must be verified by an authorized Credential Service Provider (CSP) representative through examination of physical documentation as described in SP 800-63A.

- Authenticator Assurance Level (AAL) refers to the authentication process.

- Federation Assurance Level (FAL) refers to the strength of an assertion in a federated environment, used to communicate authentication and attribute information (if applicable) to a relying party (RP).